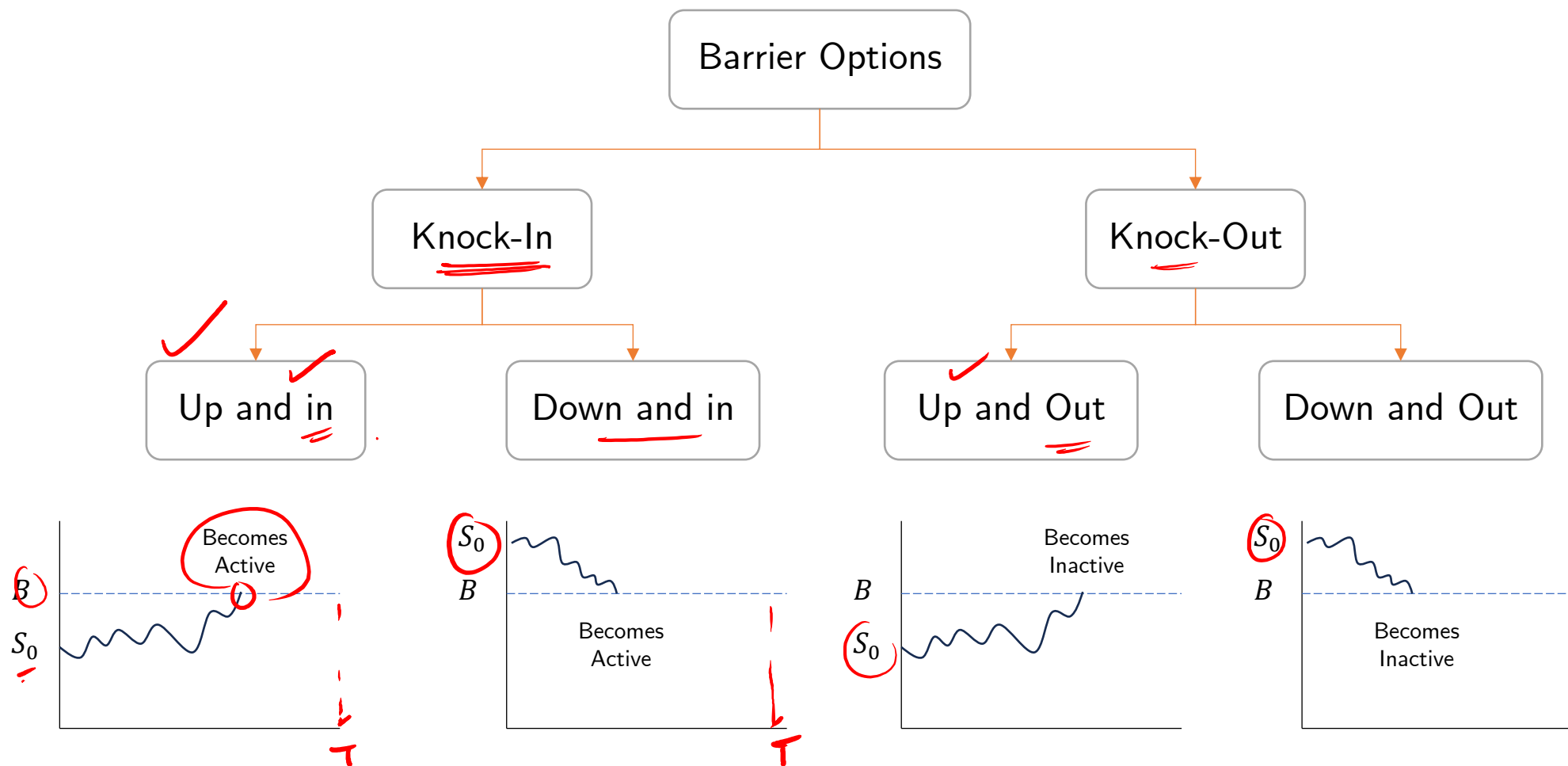


Barrier Options (2. Analytical Solutions)

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Barrier Options

Barrier options come in the following flavors



In-Out Parity

Just as we have put-call parity for European Options, we have in-out parity for Barrier Options. The parity states that a portfolio of European **knock-in** and **knock-out** options matches the **Vanilla** option.

up-and-out
call

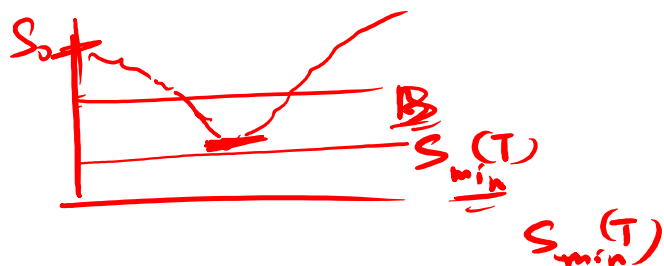
$$\rightarrow \underline{V_{k/i}(S_0, K, B, T)} + \underline{V_{k/o}(S_0, K, B, T)} = \underline{V_{bs}(S_0, K, T)} \quad \checkmark$$

The proof is straightforward. At the maturity, only one of the barrier options will be active which will match the vanilla payoff. For example, consider **call** option (similarly put)

$$(S_T - K)^+ = \max(S_T - K, 0)$$

$$(K - S_T)^+$$

$$\begin{aligned} & \downarrow \textcircled{1} \checkmark \\ & \underline{(S_T - K)^+ 1_{S_T^{max} \geq B}} + \underline{(S_T - K)^+ 1_{S_T^{max} < B}} = \underline{(S_T - K)^+} \\ & \quad \text{(up-and-in)} \quad \text{(up-and-out)} \end{aligned}$$



$$\begin{aligned} & \underline{(S_T - K)^+ 1_{S_T^{min} \leq B}} + \underline{(S_T - K)^+ 1_{S_T^{min} > B}} = \underline{(S_T - K)^+} \\ & \quad \text{(down-and-in)} \quad \text{(down-and-out)} \end{aligned}$$

The up-and-out call(II)

To price the up-and-out call, let's start with the risk neutral dynamics $\tilde{\mathbb{P}}$ of the stock

(GBM)
$$dS(t) = rS(t)dt + \sigma S(t)dW(t)$$

The solution to the SDE is given as

$$S(t) = S(0)e^{\sigma W(t) + (r - \frac{1}{2}\sigma^2)t} = S(0)e^{\sigma \hat{W}(t)}$$

$$\sigma W + (r - \frac{1}{2}\sigma^2)t = \sigma \hat{W}$$

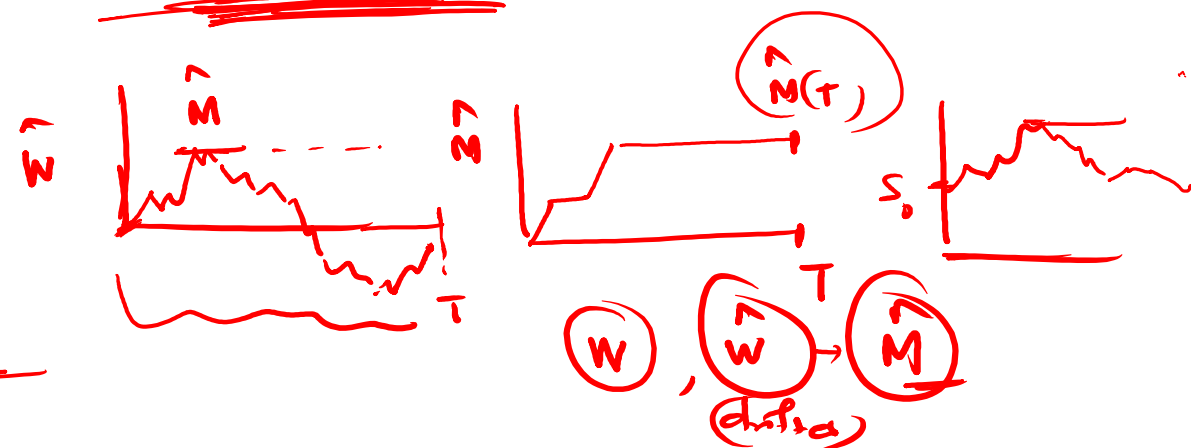
$$\Rightarrow \hat{W} = W + \frac{1}{\sigma}(r - \frac{1}{2}\sigma^2)t$$

$$= W + \alpha t$$

where $\hat{W}(t) = \alpha t + W(t)$, and $\alpha = \frac{1}{\sigma}\left(r - \frac{1}{2}\sigma^2\right)$

Also define $\hat{M}(T) = \max_{0 \leq t \leq T} \hat{W}(t)$, so $\max_{0 \leq t \leq T} S(t) = S(0)e^{\sigma \hat{M}(T)}$

The option knocks out if and only if $S(0)e^{\sigma \hat{M}(T)} > B$, else it pays off $(S_T - K)^+ = (S(0)e^{\sigma \hat{W}(T)} - K)^+$



The up-and-out call(III)

The payoff function is given as

$$\begin{aligned}
 V(T) &= \left(\overbrace{S(0)e^{\sigma\hat{W}(T)}}^{S_T} - K \right)^+ 1_{\{S(0)e^{\sigma\hat{M}(T)} \leq B\}} \\
 &= \left(S(0)e^{\sigma\hat{W}(T)} - K \right) 1_{\left\{ \underbrace{S(0)e^{\sigma\hat{W}(T)} \geq K}_{\text{(land in the money)}}, \underbrace{S(0)e^{\sigma\hat{M}(T)} \leq B}_{\text{(never crossed the barrier)}} \right\}} \rightarrow \begin{matrix} \boxed{\hat{W}(T)} \rightarrow S(T) \\ \boxed{\hat{M}(T)} \rightarrow S_{\text{barrier}}(T) \end{matrix}
 \end{aligned}$$

$S_T > K$
 brownian stock

$$= \left(S(0)e^{\sigma\hat{W}(T)} - K \right) 1_{\{\hat{W}(T) \geq k, \hat{M}(T) \leq b\}}$$

where

$$k = \frac{1}{\sigma} \log\left(\frac{K}{S(0)}\right) \quad b = \frac{1}{\sigma} \log\left(\frac{B}{S(0)}\right)$$

$$\begin{aligned}
 S(0)e^{\sigma\hat{W}(T)} &\geq K \\
 \Rightarrow e^{\sigma\hat{W}(T)} &\geq K/S(0) \\
 \Rightarrow \hat{W}(T) &\geq \frac{1}{\sigma} \log\left(\frac{K}{S(0)}\right) \\
 &\quad \underbrace{\hspace{10em}}_k \\
 \hat{W}(T) &\geq k
 \end{aligned}$$

The up-and-out call(IV)

The price today is given as

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$V(0) = \mathbb{E}^{\tilde{\mathbb{P}}} [e^{-rT} (S(0)e^{\sigma \hat{W}(T)} - K) 1_{\{\hat{W}(T) \geq k, \hat{M}(T) \leq b\}}]$$

$$\hat{W}(T) = \alpha t + W$$

$$\hat{W}, \hat{M} \leftarrow$$

To solve the above, we need the joint density between $(\hat{W}(T), \hat{M}(T))$ which is Brownian motion with drift and its running maximum. To derive the joint density, we will use the concept of Reflection equality.

First let's consider the simple case where we have the standard Brownian motion $W(t)$. We will show that the joint density of the standard Brownian and its running maximum is given as

$$g = g(x, y)$$

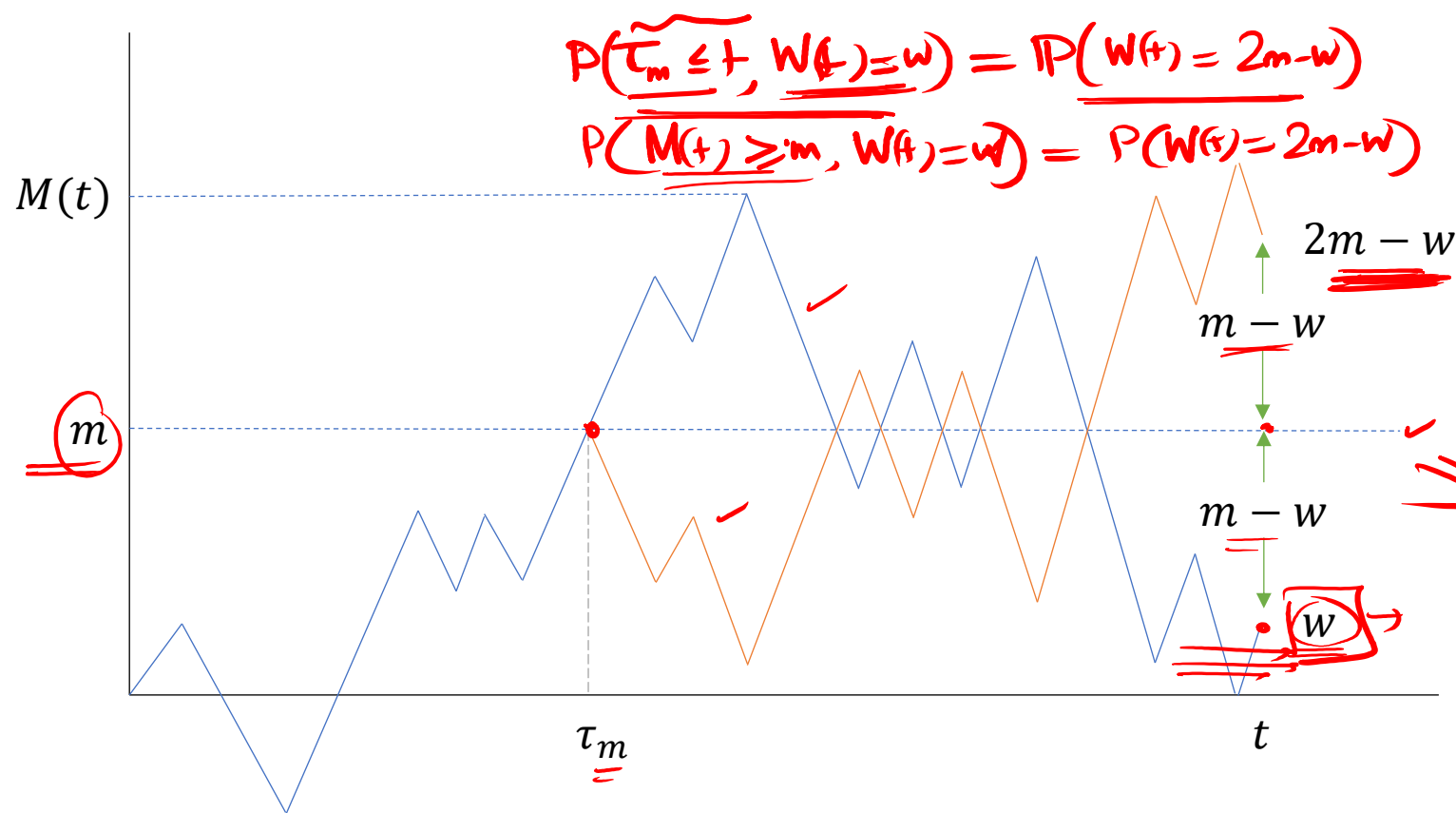
$$\mathbb{E}(g) = \iint_{-\infty}^{\infty} g f(x, y) dx dy$$

$$f_{M(t), W(t)}(m, w) = \frac{2(2m - w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

$$\underline{W} \rightarrow \underline{M}$$

First passage time distribution(I)

What is the probability of the standard Brownian W crossing barrier m before t and reaching a level w at time t ? Let's denote $\tau_m = \min\{s \geq 0, W(s) = m\}$ as the First Passage time for hitting the barrier at m . Consider the reflected path around the barrier after the process hits the barrier.



All paths that cross the barrier(m) before t and reach w at t , have a bijection between the paths that reach $(2m-w)$ at t . Also, paths that end up below w have the reflected path end above $(2m-w)$. So, we can write

$$\mathbb{P}\{\tau_m \leq t, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$$

(Reflection Equality)

First passage time distribution(II)

Substitute $w = m$, in the equation to get

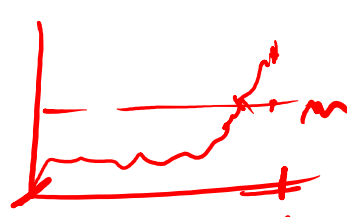
$$\mathbb{P}\{\tau_m \leq t, W(t) \leq m\} = \mathbb{P}\{W(t) \geq m\}$$

The CDF of the first passage time τ_m can be written as

$$\mathbb{P}\{\tau_m \leq t\} = \mathbb{P}\{\tau_m \leq t, W(t) \leq m\} + \mathbb{P}\{\tau_m \leq t, W(t) \geq m\}$$

$$= \mathbb{P}\{W(t) \geq m\} + \mathbb{P}\{W(t) \geq m\}$$

$$= 2\mathbb{P}\{W(t) \geq m\} = \frac{2}{\sqrt{2\pi t}} \int_m^{\infty} e^{-\frac{x^2}{2t}} dx$$



$$W(t) \sim N(0, t)$$

$$\frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

With variable substitution $y = \frac{x}{\sqrt{t}}$, we get $\mathbb{P}\{\tau_m \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|m|}{\sqrt{t}}}^{\infty} e^{-\frac{y^2}{2}} dy$

And the density $f_{\tau_m}(t) = \frac{d}{dt} \mathbb{P}\{\tau_m \leq t\} = \frac{|m|}{t\sqrt{2\pi t}} e^{-\frac{m^2}{2t}}$

$y = x/\sqrt{t}$
 $dy = \frac{1}{\sqrt{t}} \cdot dx$

$$P(\tau_m \leq t) = \frac{2}{\sqrt{2\pi t}} \int_{\frac{|m|}{\sqrt{t}}}^{\infty} e^{-\frac{y^2}{2}} dy$$

Joint Distribution of Brownian and Max (W, M)

If W is the Brownian motion and M is the running max $M(t) = \max_{0 \leq s \leq t} W(s)$, from the reflection equality,

$$\begin{aligned} \mathbb{P}\{\tau_m \leq t, W(t) \leq w\} &= \mathbb{P}\{W(t) \geq 2m - w\} \quad \leftarrow \\ \Rightarrow \mathbb{P}\{M(t) \geq m, W(t) \leq w\} &= \mathbb{P}\{W(t) \geq 2m - w\} \quad W(t) \sim N(0, t) \\ \Rightarrow \int_m^\infty \int_{-\infty}^w f_{M,W}(x, y) dy dx &= \frac{1}{\sqrt{2\pi t}} \int_{2m-w}^\infty e^{-\frac{z^2}{2t}} dz \end{aligned}$$

Differentiate w.r.t m , we get

$$-\int_{-\infty}^w f_{M,W}(m, y) dy = -\frac{2}{\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

Next, we differentiate w.r.t w , we get

$$\left(\frac{\partial}{\partial w}, \frac{\partial}{\partial m} \right) \frac{\partial}{\partial w} = \frac{\partial}{\partial w} \left(\frac{\partial}{\partial m} + \frac{\partial}{\partial w} \right) \Rightarrow \frac{\partial}{\partial w} f_{M,W}(m, w) = \frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

$f_{(M, W)} = f_{m, w}$

Joint Distribution of drifted Brownian and Max $(\hat{W}, \hat{M})$

Now suppose we have the Brownian with drift and its running max as we started with

$$ds = rSdt + \sigma SdW \quad \text{②} \quad \hat{W}(t) = \alpha t + W(t) \quad \hat{M}(T) = \max_{0 \leq t \leq T} \hat{W}(t) \quad \hat{W} = -\alpha t + W$$

Now, W is a standard Brownian under $\tilde{\mathbb{P}}$. Consider a measure $\tilde{\mathbb{P}}$, under which $\hat{W}$ is the standard Brownian. The measures are related through Radon Nikodym derivative

$$\tilde{\mathbb{P}} \rightarrow \hat{W} \xrightarrow{\text{SBM}} \hat{W} \quad \hat{Z}(t) = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = \exp\left(-\alpha W(t) - \frac{1}{2}\alpha^2 t\right) = \exp\left(-\alpha \hat{W}(t) + \frac{1}{2}\alpha^2 t\right)$$

$W = \hat{W} - \alpha t$
 $-\alpha W = -\alpha(\hat{W} - \alpha t) = -\alpha \hat{W} + \alpha^2 t$

To get the joint density of $(\hat{W}, \hat{M})$ under $\tilde{\mathbb{P}}$, we write

$$\tilde{\mathbb{P}}[\hat{M}(T) \leq m, \hat{W}(T) \leq w] = \mathbb{E}^{\tilde{\mathbb{P}}}[1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}}] = \int x d\tilde{\mathbb{P}} = \int (x \frac{1}{\hat{Z}(T)}) d\mathbb{P} = \mathbb{E}^{\mathbb{P}}\left[\frac{x}{\hat{Z}(T)} 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}}\right]$$

CDF
 $\mathbb{E}(1_{S > k})$
 $S > k, \frac{1}{2}$
 $S < k, 0$
 $\frac{1}{2} \times P(S > k) + 0 \times P(S < k)$

Joint Distribution of drifted Brownian and Max $(\hat{W}, \hat{M})$

$$\begin{aligned}
 \tilde{\mathbb{P}}\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\} &= \hat{\mathbb{E}} \left[\frac{1}{\hat{Z}(T)} 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}} \right] \\
 &= \hat{\mathbb{E}} \left[\exp \left(\alpha \hat{W}(t) - \frac{1}{2} \alpha^2 t \right) 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}} \right] \\
 &= \int_{-\infty}^w \int_{-\infty}^m e^{\alpha y - \frac{1}{2} \alpha^2 T} \hat{f}_{\hat{M}, \hat{W}}(x, y) dx dy
 \end{aligned}$$

Differentiating twice, we recover the joint density under $\tilde{\mathbb{P}}$

$$\frac{\partial^2}{\partial m \partial w} \tilde{\mathbb{P}}\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\} = e^{\alpha w - \frac{1}{2} \alpha^2 T} \hat{f}_{\hat{M}, \hat{W}}(m, w)$$

$$\hat{f}_{\hat{M}, \hat{W}}(m, w) = e^{\alpha w - \frac{1}{2} \alpha^2 T} \frac{2(2m - w)}{t \sqrt{2\pi t}} e^{-\frac{(2m - w)^2}{2t}}$$

Corollary

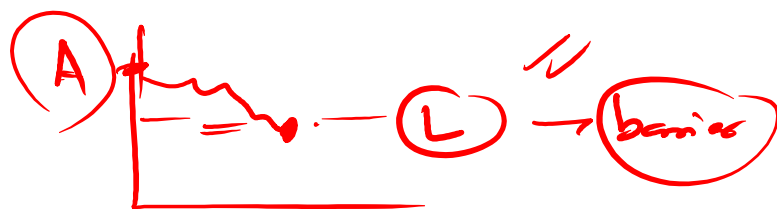


$\hat{M} \leftarrow \hat{W} \otimes \alpha t$

$$\tilde{\mathbb{P}}\{\hat{M}(T) \leq m\} = N\left(\frac{m - \alpha T}{\sqrt{T}}\right) - e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right)$$

And the density under $\tilde{\mathbb{P}}$ of the random variable $\hat{M}(T)$ is

$$\underline{\underline{\tilde{f}_{\hat{M}(t)}}(m)} = \frac{2}{\sqrt{2\pi T}} e^{-\frac{1}{2T}(m - \alpha T)^2} - 2\alpha e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right)$$



(Up-and-Out) Barrier Option

We have everything we need to evaluate the final integral which is the expected discounted price

$$\begin{aligned}
 \underline{\underline{V(0)}} &= \underline{\underline{\mathbb{E}^{\tilde{\mathbb{P}}}}} \left[\underline{\underline{e^{-rT}}} \left(\underline{\underline{S(0)}} \underline{\underline{e^{\hat{W}(t)}}} - \underline{\underline{K}} \right) \underline{\underline{1_{\{\hat{W}(T) \geq k, \hat{M}(T) \leq b\}}}} \right] \\
 &= \int_k^b \int_w^b \underline{\underline{e^{-rT}}} \left(\underline{\underline{S(0)}} \underline{\underline{e^{\hat{W}(t)}}} - \underline{\underline{K}} \right) \underline{\underline{e^{\alpha w - \frac{1}{2}\alpha^2 T}}} \frac{2(2m - w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}} dm dw
 \end{aligned}$$

$k = \frac{1}{\sigma} \log\left(\frac{K}{S_0}\right)$
 $b = \frac{1}{\sigma} \log\left(\frac{B}{S_0}\right)$

After simplification, we get

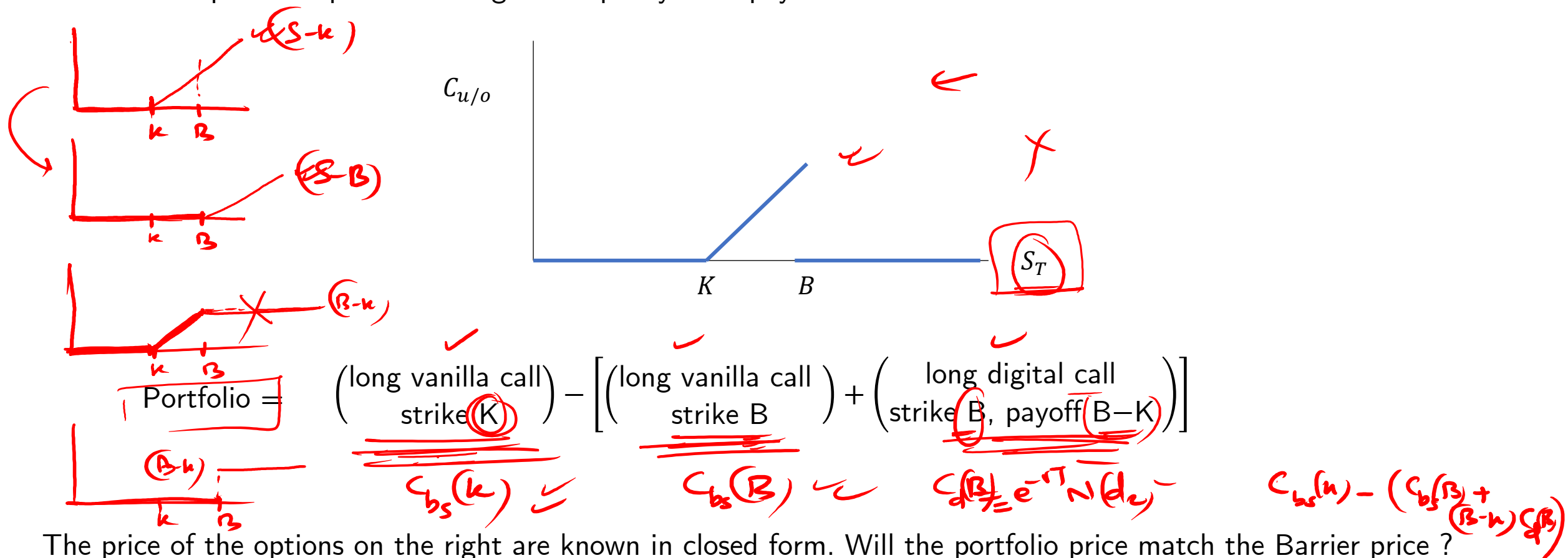
$$\underline{\underline{C_{u/o}(S_0, K, B, T)}} = \left\{ \underline{\underline{C_{bs}(S_0, K, T)}} - \left[\underline{\underline{C_{bs}(S_0, B, T)}} + (B - K) \underline{\underline{C_d(S_0, B, T)}} \right] \right\} - \left(\frac{S_0}{B} \right)^{2\lambda} \left\{ \underline{\underline{C_{bs}\left(\frac{B^2}{S_0}, K, T\right)}} - \left[\underline{\underline{C_{bs}\left(\frac{B^2}{S_0}, B, T\right)}} + (B - K) \underline{\underline{C_d\left(\frac{B^2}{S_0}, B, T\right)}} \right] \right\}$$

Where C_{bs} is vanilla BSM call price and C_d is the digital call price and $\lambda = \frac{1}{2} \left[1 - \frac{r}{\frac{1}{2}\sigma^2} \right]$



The up-and-out call(I)

To start off, let's focus on the **up-and-out** call option. If we know the price of the up-and-out, we can immediately calculate the price of up-and-in using in-out parity. The payoff is discontinuous



The price of the options on the right are known in closed form. Will the portfolio price match the Barrier price ?

Barrier Option Formulae

Call

- Up-and-out** →
- Up-and-in**
- Down-and-out**
- Down-and-in**

$$C_{u/o}(S_0, K, B, T) = C_{bs}(S_0, K, T) - [C_{bs}(S_0, B, T) + (B - K)C_d(S_0, B, T)] - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ C_{bs}\left(\frac{B^2}{S_0}, K, T\right) - \left[C_{bs}\left(\frac{B^2}{S_0}, B, T\right) + (B - K)C_d\left(\frac{B^2}{S_0}, B, T\right) \right] \right\}$$

$$C_{u/i}(S_0, K, B, T) = C_{bs}(S_0, K, T) - C_{u/o}(S_0, K, B, T)$$

$$C_{d/o}(S_0, K, B, T) = C_{bs}(S_0, Z, T) + (Z - K)C_d(S_0, Z, T) - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ C_{bs}\left(\frac{B^2}{S_0}, Z, T\right) + (Z - K)C_d\left(\frac{B^2}{S_0}, Z, T\right) \right\} \quad Z = \max\{B, K\}$$

$$C_{d/i}(S_0, K, B, T) = C_{bs}(S_0, K, T) - C_{d/o}(S_0, K, B, T)$$

Put

- Up-and-out**
- Up-and-in**
- Down-and-out**
- Down-and-in**

$$P_{u/o}(S_0, K, B, T) = P_{bs}(S_0, Z, T) + (K - Z)P_d(S_0, Z, T) - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ P_{bs}\left(\frac{B^2}{S_0}, Z, T\right) + (K - Z)P_d\left(\frac{B^2}{S_0}, Z, T\right) \right\} \quad Z = \max\{B, K\}$$

$$P_{u/i}(S_0, K, B, T) = P_{bs}(S_0, K, T) - P_{u/o}(S_0, K, B, T)$$

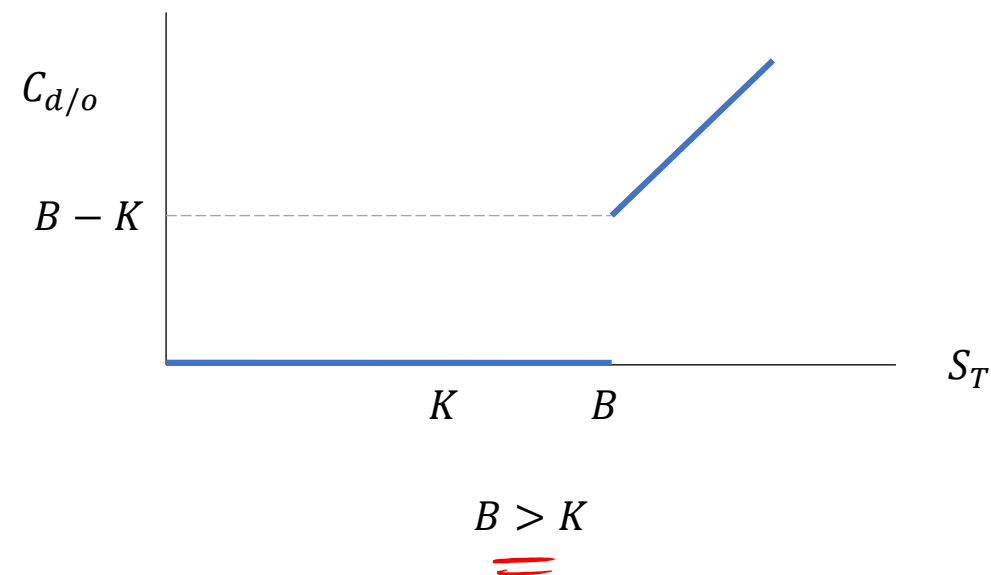
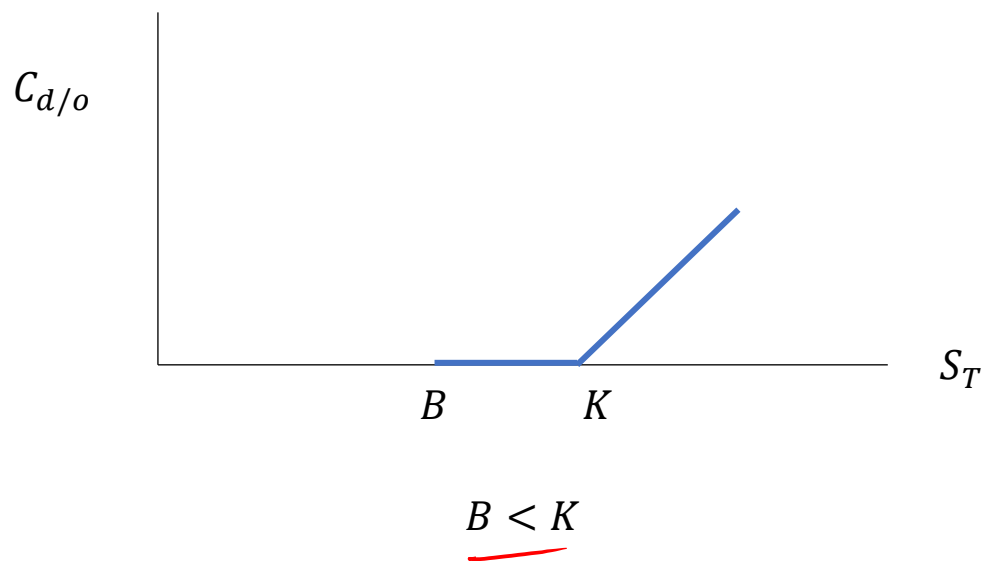
$$P_{d/o}(S_0, K, B, T) = P_{bs}(S_0, K, T) - [P_{bs}(S_0, B, T) + (K - B)P_d(S_0, B, T)] - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ P_{bs}\left(\frac{B^2}{S_0}, K, T\right) - \left[P_{bs}\left(\frac{B^2}{S_0}, B, T\right) + (K - B)P_d\left(\frac{B^2}{S_0}, B, T\right) \right] \right\}$$

$$P_{d/i}(S_0, K, B, T) = P_{bs}(S_0, K, T) - P_{d/o}(S_0, K, B, T)$$

Exercise

What is the payoff diagram for Down-and-Out Call ?

What is the synthetic portfolio that mirrors the pay-off of Down-and-out call ?



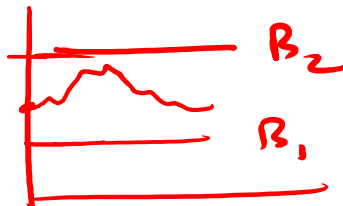
Variants



Barrier options with rebates...

Double barrier... ✓

American barrier...



Homework



1. Up-and-out Put Barrier Option.
 - a) What is the payoff diagram ?
 - b) What is the Synthetic portfolio that mirrors the payoff ?
 - c) Find MCS price and compare against analytical solution.